

APOLLO 13	
LM SYSTEMS ACTIVATION CHECKLIST	
PART NO.	S / N



NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

APOLLO XIII LM-7

ACTIVATION CHECKLIST

PREPARED BY
FLIGHT CREW SUPPORT DIVISION
SPACECRAFT SYSTEMS BRANCH

MANNED SPACECRAFT CENTER
HOUSTON, TEXAS

FEBRUARY 6, 1970

REVISED MARCH 25, 1970



APOLLO 13

ACTIVATION CHECKLIST

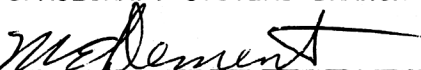
February 6, 1970
REVISED MARCH 25, 1970

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APOLLO FLIGHT DATA FILE

CHANGE DATE 3/25/70

LIST OF EFFECTIVE PAGES

Basic Date _____
Changed _____

<u>PAGE NO.</u>	<u>BASIC DATE</u>	<u>CHANGED DATE</u>
COVER	2/6/70	3/25/70
SIGNATURE PAGE	2/6/70	3/25/70
CSM to LM Transfer (TLC).	2/6/70	3/16/70
TLC 1 and 2	2/6/70	NONE
ACT 1 thru 3	2/6/70	NONE
ACT 4	2/6/70	3/16/70
ACT 5 thru 9	2/6/70	NONE
ACT 10	2/6/70	3/16/70
ACT 11	2/6/70	3/25/70
ACT 12 thru 14	2/6/70	NONE
ACT 15	2/6/70	3/16/70
ACT 16	2/6/70	NONE
CSM to LM Transfer (PDI).	2/6/70	NONE
ACT 17 and 18	2/6/70	3/16/70
ACT 19 thru 21	2/6/70	NONE
ACT 22	2/6/70	3/16/70
ACT 23 thru 26	2/6/70	NONE
ACT 27 and 28	2/6/70	3/16/70
ACT 29 thru 33	2/6/70	NONE
ACT 34	2/6/70	3/16/70
ACT 35 thru 40	2/6/70	NONE
ACT 41 and 42	2/6/70	3/16/70
ACT 43 thru 49	2/6/70	NONE
ACT 50	2/6/70	3/25/70
ACT 51	2/6/70	3/16/70
ACT 52 thru 56	2/6/70	NONE
ACT 57	2/6/70	3/16/70
ACT 58 and 59	2/6/70	NONE

Basic Date _____ 2/6/70
Changed _____ 3/16/70

CSM TO LM TRANSFER LIST(TLC)

Tissue Dispenser - RHSSC

16MM LS Camera

16MM Magazines (8)

70MM Magazines in Bag (3) - RHSSC

70MM Magazines HCEX (2) - ISA

SUIT HOSE INTERCONNECT

VACUUM HOSE (3 ft)

VACUUM BRUSH

CWG CONNECTOR

Flight Data In Bag:

LM ACTIVATION CHECKLISTS (2)

Basic Date 2/6/70
Changed
TLC-1

58:00

IVT TO LM

- 1 Activate CABIN DUMP VALVE & Open Hatch
 Take CSM 02 Hose When Transferring
- 2 Record Docking Tunnel Index Angle
 Rc
- 3 FLOOD LIGHT - A11
 EXTERIOR LTG - OFF
 Window Shades - Down
- 4 DES H20 - OPEN
 DES 02 - OPEN
 CABIN REPRESS - AUTO
 CB(16) CABIN REPRESS - CLOSE
- 5 Check AOT Visibility

IVT TO LM

TLC-2

59:45

IVT TO CSM

- 1 DES H20 - CLOSE
DES 02 - CLOSE
CABIN REPRESS - CLOSE
CB(16) CABIN REPRESS - Open
FLOOD LIGHT - OFF
Window Shades - Up
- 2 CABIN DUMP VALVE - OPEN
IVT TO CSM
CLOSE LM HATCH

LM 7

Basic Date 2/6/70
Changed

Basic Date _____ 2/6/70
Changed _____

	LOI DAY	
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LOI DAY

Basic Date 2/6/70

Changed

ACT-1

83:09LMP IVT TO LM

- 1 Activate CABIN DUMP VALVE & Open Hatch
Carry Comm Carrier, CWG Connector &
CSM 02 Hose
- 2 FLOOD LIGHT - All
Window Shades - Down
- 3 DES H2O - OPEN
DES 02 - OPEN
CABIN REPRESS - AUTO
CB(16) CABIN REPRESS - Close

LOS 83:13

ACT-2

83:14ENTRY STATUS CHECK

- 1 Unstow Purse And Give To CSM
Unstow ISA And Install Over PLSS
Recharge Station
- 2 Verify CB Status Per INITIAL ACTIVATION
Status Chart

ACT-3

[illegible]

INITIAL ACTIVATION STATUS

[illegible]

LIT		ID		STAM/CONT				INST		ECS	
FLOOD	TRACE	ARMED/ALARM	MAINTENANCE	ALA	ARM	ARM	ARM	ARM	ARM	ARM	ARM

COMMA							ECS									
SMP	LI AUDIO	NIP A MATE	NIP B MCVIA	VIDEO SUBTITLE PREF JUMP	LSD ANT	FMP	TV	DSP	OUTSIDE PUMP SEC	ISC PUMP	CARDIAN PUMP CONT	PUMP FAN 2	PUMP FAN 1 P	PUMP FAN 1 V1	CDI SENSOR	

HEATERS			CAMM			EPS			FIBERGLASS			NOT FIELD USE		
QUAD 1	QUAD 2	QUAD 3	DISP	SEC	SHO	DISP	DISP	DISP	DISP	DISP	DISP	DISP	DISP	DISP
			<											

Basic Date _____ 2/6/70
Changed _____

ACT-5

- 3 RR GYRO SEL - PRIM
- 4 FDAI 1&2 - INRTL
EARTH/LUNAR - PWR OFF
LTG - OFF
MODE - HOLD/FAST
ALT SET - 60
- 5 DES PROP ISOL - SAFE
MASTER ARM - OFF
ASC He SEL - BOTH
STAGE - SAFE (Guarded)
- 6 S-BAND T/R - OFF
ICS T/R - OFF
RELAY - OFF
MODE - ICS/PTT
AUDIO CONT - NORM
VHF A&B - OFF
VOX SENS - 9
COAS - OFF
THUMBWHEEL VOL (5)-6
- 7 TTCA (CDR) - JETS

ACT-6

8

TIMER CONT - STOP
LTG OVERRIDE (3) - OFF
SIDE PANELS - OFF
FLOOD OVHD/FWD - BRIGHT
ANUN/NUM - DIM
INTEGRAL - DIM

9

X-POINTER SCALE - HI MULT
RATE/ERR MON - LDG RDR/CMPTR
ATTITUDE MON - PGNS
GUID CONT - PGNS
MODE SEL - LDG RADAR
RNG/ALT MON - ALT/ALT RT
SHFT/TRUN - +50°
RATE SCALE - 25°/SEC
ACA PROP - ENABLE
THR CONT - AUTO
MAN THROT - CDR
ENG ARM - OFF
ATT/TRANSL - 2 JETS
BAL CPL - ON
ASC He REG 1&2 - tb-gray (v1v Open)
DESCENT He REG 1-tb-gray (v1v Open)
DESCENT He REG 2-tb-bp (v1v Closed)
PRPLNT QTY MON - OFF
PRPLNT TEMP/PRESS MON - ASC
HELIUM MON - OFF
ABORT and ABORT STAGE - Flush/Guarded

LM 7

Basic Date 2/6/70

Changed

Basic Date _____ 2/6/70

Changed _____

ACT-7

- 10 SYS A&B ASC FUEL & ASC OXID(4) -tb-bp
 (Feed 2-Close, Feed 1-Open)
 SYS A&B QUADS (8)-tb-gray (v1v open)
 CRSFD -tb-bp (v1v closed)
 SYS A&B MAIN SOV - tb-gray (v1v open)
 TEMP/PRESS MON - He
 ACA PROP - DISABLE
 RATE/ERR MON - LDG RDR/CMPTR
 ATTITUDE MON - AGS
 GLYCOL - PUMP 2
 SUIT FAN - 1
 O2/H2O QTY MON - ASC 2
- 11 ENG GMBL - ENABLE
 DES ENG CMD OVRD - OFF
 LDG ANT - AUTO
 RADAR TEST - OFF
 TEST MONITOR - ALT XMTR
 SLEW RATE - HI
 RNDZ RDR - SLEW
 DEAD BAND - MIN
 GYRO TEST - ROLL
 ATTITUDE CONTROL (3) - MODE CONT
 MODE CONT: (Both) - OFF
 EVENT TIMER: TIMER CONT - STOP
 TEMP MON - LDG
 RCS SYS A/B-2 QUADS - OFF
 LTG: SIDE PANELS - OFF
 FLOOD-A11

OVHD/FWD - BRIGHT
 EXTERIOR LTG - OFF
 LAMP/TONE TEST - OFF
 X-POINTER SCALE - HI MULT

12

ACA/4 JET (CDR) - ENABLE
 ACA/4 JET (LMP) - DISABLE
 TTCA/TRANSL (CDR) - ENABLE
 TTCA/TRANSL (LMP) - DISABLE
 RNDZ RDR ANT - Stowed
 AOT - CL, ANGLE - 0000 (Pushed In)
 TTCA (LMP) - JETS
 AGS STATUS - OFF

13

PWR TEMP MON-ED/OFF
 INV-OFF
 DES PWR (5)-tb-bp
 ASC PWR (4)-tb-bp
 UPLINK SQUELCH-ENABLE

14

AUDIO CONT - NORM
 S-BAND T/R - OFF
 ICS T/R - OFF
 RELAY - OFF
 MODE - ICS/PTT
 UPDATA LINK - OFF
 VHF A&B - OFF
 VOX SENS - 9
 THUMBWHEEL VOL(5)-6

Basic Date — 2/6/70
 Changed —

Basic Date _____ 2/6/70
 Changed _____

ACT-9

15

S-BAND MODULATE - PM
 XMTR/RCVR - OFF
 PWR AMPL - OFF
 VOICE - OFF
 PCM - OFF
 RANGE - OFF/RESET
 VHF A - OFF (SQUELCH-3)
 VHF B - OFF (SQUELCH-3)
 TELEMETRY - OFF/HI
 RECORDER - OFF
 VHF - AFT
 TRACK MODE - OFF
 PITCH - -75°
 YAW - -12°
 S-BAND - AFT

16

SUIT GAS DIVERTER - PULL/EGRESS
 CABIN REPRESS - AUTO
 PLSS FILL - CLOSE
 PRESS REG A&B - CLOSE
 DES 02 - OPEN
 ASC 02(2)-CLOSE
 SUIT ISOL (2) - SUIT DISC
 SUIT CIRCUIT RELIEF - AUTO
 CABIN GAS RETURN - AUTO
 CO2 CANISTER SEL - PRIM
 PRIM & SEC CO2 CANISTER - CLOSE
 WATER SEP SEL - PULL/SEP 2
 ASC H2O - CLOSE

ACT-10

SEC EVAP FLOW - CLOSE
PRIM EVAP FLOW (2)-CLOSE
DES H2O-OPEN
WATER TANK SELECT -DES
SUIT TEMP - COLD
LIQUID COOLING GARMENT - COLD

17 Verify (192 PKG) Lanyard
Not Seated

18 FWD CABIN RELIEF AND DUMP - AUTO

83:19

HOUSEKEEPING

1 Unsnap LMP's HSB And Stow Next To
CDR's HSB On Floor Velcro. Unsnap
CDR's HSB

2 Unstow 70mm Film Bag (Top Left
of RHSSC)
Put Up Bungee Straps

3 Install Pn1 16 Qd 1 HTR CB
Guard (FDF Acc. Kit)

Basic Date 2/6/70
Changed 3/16/70

ACT-17

4 Insta11 16mm CAMR Wedge (LS CAMR Bag)

[illegible]

SR 83:37

A05 84.07

A0S 84:01

84:17

COMM ACTIVATION

- 1 Transfer To LM POWER (FLOOD Lts. Blink,
C/W PWR Caution Lt - On)
GET _____ : (Report To MCC)
CB(11) EPS: XLUNAR BUS TIE - Close
CB(16) EPS: XLUNAR BUS TIE - Close
CB(11) LTG: UTIL - Close
Activate Utility Lights
- 2 CB(11) COMM: VHF B XMTR - Close
 : VHF A RCVR - Close
 : CDR AUDIO - Close
 INST: SIG CONDR 1- Close
 ECS: GLYCOL PUMP 2- Close

ACT-12

- 3 CB(16) INST: SIG CONDR 2-Close
 EPS: DISP - Close
 : DES ECA CONT-Close
 Verify DES BATS tb(4) - LO,DES BATS-tb
 gray
 PWR/TEMP MON - Check Voltages

When BUS VOLT < 27V, Select HI Voltage Taps
 CB(16)EPS: CROSS TIE BAL LOADS - Open
 BAT 1 HI VOLTAGE - OFF/RESET
 BAT 1 HI VOLTAGE - ON
 Repeat For BATS 2,3,4
 CB(16) EPS: CROSS TIE BAL LOADS - Close

- 4 CB(16) COMM: DISP - Close
 : VHF A XMTR - Close
 : VHF B RCVR - Close
 : PRIM S-BD (2) - Close
 : PMP - Close
 INST: SIG SENSOR - Close
 : PCM/TE - Close
 ECS: DISP - Close
- 5 Connect To LM COMM Umbilical Using
 CWG Connector
- 6 CB(16) SE AUDIO - Close

Basic Date _____ 2/6/70
 Changed _____

Basic Date _____ 2/6/70
 Changed _____
 ACT-13

84:29

* S-BAND/VHF SIMPLEX VOICE TEST

- 1 AUDIO (LMP): S-BAND T/R - T/R
 - : VHF A - T/R
 - : VHF B - OFF

COMM: S-BAND-PM, PRIM, PRIM, DN VOICE BU,
 PCM, OFF/RESET, OFF, LO

 - VHF A XMTR - VOICE
 - VHF A RCVR - ON
 - S-BAND ANT - AFT

Perform VHF A Voice Check With CSM
- 2 COMM: VHF A XMTR & RCVR - OFF
 - : VHF B XMTR - VOICE
 - : VHF B RCVR - ON

AUDIO (LMP): VHF A-OFF

 - : VHF B-T/R

Perform VHF B Voice Check With CSM
- 3 Perform S-BD Voice & LBR Check With MSFN
 TLM-HI
 Perform Voice & HBR Check With MSFN
- 4 BIOMED-RIGHT
 Perform Voice & HBR Check With MSFN

ACT-14

- 5 TLM-LO
Perform Voice & LBR Check With MSFN
- 6 S-BAND: VOICE-VOICE
Perform Voice & LBR Check With MSFN
- 7 TLM-HI
Perform Voice & HBR Check With MSFN
- 8 TLM-LO
S-BAND: RANGE-RANGE
Perform Voice & Ranging Check With MSFN
- 9 CB(16) CAMR: SEQ - Close
Check SEQ Camera Operation

SS 84:42

84:44

OPS CHECKOUT

- 1 Perform OPS Checkout
Read And Record Source Pressures
CDR OPS _____

2/6/70

Basic Date _____

Changed _____

LMP OPS _____

LM 7

Basic Date _____ 2/6/70
Changed _____ 3/16/70
ACT-15

84:49

COMM DEACTIVATION

- 1 AUDIO (LMP): S-BAND T/R - OFF
: VHF B - OFF
- 2 COMM: S-BAND - PM,OFF,OFF,OFF,OFF,
OFF/RESET,OFF,L0
: VHF B XMTR - OFF
: VHF B RCVR - OFF
- 3 CB(16)EPS: CROSS TIE BAL LOADS-Open
Select L0 TAPS
UTILITY LTS - OFF
- 4 Configure CB Panels Per INT ACT STATUS
Chart (ACT 3,4)
Disconnect From LM Comm Umbilical
- 5 Transfer To CSM Power, Observe C/W
PWR Lt - Off
GET ____:____:____ (Report To MCC) |

IVT TO CSM

IVT TO CSM

ACT-16

85:00

LMP IVT TO CSM

1 DES 02 - CLOSE
DES H20 - CLOSE
CABIN REPRESS - CLOSE
CB(16) ECS: CABIN REPRESS - Open
Window Shades - Up

2 FLOOD LIGHT - OFF
Check AOT Visibility

3 CABIN RELIEF & DUMP (OVHD) - Open
IVT TO CSM, Close LM Hatch

LOS 85:06 *****

SR 85:31 *****

LM 7

Basic Date 2/6/70
Changed

Basic Date _____ 2/6/70
Changed _____

PDI DAY	

PDI DAY

LM 7

Basic Date _____ 2/6/70
Changed _____

Basic Date _____ 2/6/70
Changed _____

CSM TO LM TRANSFER LIST(PDI)

Suits And Ancillary Eqpt:

Personal Radiation Dosimeter

Liners

Flight Data In Bag:

IV Gloves

LM TIMELINE BOOK

Helmet

LM DATA CARD BOOK

Bio belt & Instrumentation

LM LUNAR SURFACE CHECKLIST

Comm Cap

LM ORBIT MONITOR CHART

Wristwatch (2)

LUNAR PHOTO CHARTS

Sunglasses in pouch

STAR CHARTS

Pens & Pencils

Penlight

Scissors

LM 7
Basic Date _____ 2/6/70
Changed _____ 3/16/70
ACT-17

96:39

LMP IVT TO LM

- 1 Activate CABIN DUMP VALVE & Open Hatch
 Carry Comm Carrier & CSM 02 Hose
 Mount Purse
- 2 Verify Docking Tunnel Index
 Angle (See TLC-1)
 Window Shades - Down
 Deploy LMP Crash Bar
- 3 Transfer To LM PWR
 GET _____ : _____
 (FLOOD Lts. Blink, C/W PWR Caution Lt-0n)
 CB(11) EPS: XLUNAR BUS TIE - Close
 CB(16) EPS: XLUNAR BUS TIE - Close
- 4 FLOOD LIGHT - All
 CB(11) LTG: UTIL - Close
 Activate Utility Lts
- 5 DES H20 - OPEN
 DES 02 - OPEN
 CABIN REPRESS - AUTO
 CB(16) ECS: CABIN REPRESS - Close

IVT TO LM
EPS ACT

ACT-18

***** UD - 2:30 (96:46) *****

***** SR 96:54 *****

96:56

CDR IVT TO LM

CDR IVT To LM With CDR &
LMP Helmet & Gloves

Connect To LM Comm Umbilical
CB(11) COMM: CDR AUDIO - Close
AUDIO (CDR): S-BAND-T/R
: ICS - T/R

96:56

EPS ACTIVATION

- | | |
|---|--|
| 1 | LTG: ANUN/NUM - BRIGHT (1 Caution, 9
Power Failure, 1 COMP Lt - On) |
| 2 | CB(11) INST: SIG CONDR 1 - Close
EPS: DES ECA CONT - Close
CB(16) INST: SIG SENSOR - Close
: PCM/TE - Close
: SIG CONDR 2 - Close
EPS: DISP - Close
: DES ECA CONT - Close |
| 3 | Connect To LM Comm Umbilical |

LM 7

Basic Date 2/6/70
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Basic Date _____ 2/6/70
 Changed _____

ACT-19

AUDIO (LMP): S-BAND T/R - T/R
 : ICS - T/R
 CB(11) COMM: SEC S-BD(2) - Close
 CB(16) COMM: DISP - Close
 : S.E. AUDIO - Close
 : S-BD ANT - Close
 : PMP - Close
 S-BAND - PM,SEC,SEC,VOICE,PCM,RANGE,
 OFF,L0
 S-BAND ANT - AFT

- 4 Verify BAT 1,2,3,4 - tb-L0
 DES BATS tb-gray
 BATS 5&6 NORMAL & BACKUP (4)-tb-bp
 Check BAT and BUS Voltages

When BUS Volts $\leq$ 27V, Select High Voltage
 Taps
 CB(16) EPS: CROSS TIE BAL LOADS - Open
 BAT 1 HI VOLTAGE-OFF/RESET
 BAT 1 HI VOLTAGE-ON
 Repeat for BATS 2,3,4

- 5 CB(11) AC BUS B&A: BUS TIE INV 2&1(4) -
 Close
 AC BUS A : AC BUS VOLT - Close
 EPS: INV 1 - Close
 CB(16) EPS: INV 2 - Close

ACT-20

- 6 POWER/TEMP MON - AC BUS
INV -1 Then 2
Verify Voltage in GREEN Band
CB(11) EPS: INV 1 - Open

97:00

MISSION TIMER ACTIVATION

- 1 CB(11) AC BUS B: NUM LTG - Close
FLIGHT Displays: MISSION TIMER-Close
Set MSN TMR On CSM Mark

97:02

PRIMARY GLYCOL LOOP ACTIVATION

- 1 CB(16) ECS: DISP - Close
GLYCOL - PUMP 1 _____ psia
- INST(SEC) _____ psia
- PUMP 2 _____
CB(11) ECS: GLYCOL PUMP AUTO TRNFR - Close
: GLYCOL PUMP 1 - Close
: GLYCOL PUMP AUTO TRNFR-Open
GLYCOL - PUMP 1
Verify Press _____ psia
CB(11) ECS: GLYCOL PUMP 2 - Close

LM 7

Basic Date 2/6/70
Changed

LM 7

Basic Date _____ 2/6/70
Changed _____
ACT 21

97:04

CAUTION/WARNING CHECKOUT

- | | | | | | | | | | | | | | | | | | | | | | | | |
|------------------------|--|--|------|------|------|------------------------|--------|---------|--------|--------|--|--------|-----|--|-----|-------------------|--|-----------|-------|--|-----------|--|--|
| 1 | CB(16) LTG: MASTER ALARM - Close
INST: CWEA - Close | | | | | | | | | | | | | | | | | | | | | | |
| | | <table border="0"><tr><td>WARN</td><td>CAUT</td><td>COMP</td></tr><tr><td>ASC PRESS(ON THRU DES)</td><td>PREAMP</td><td>H2O SEP</td></tr><tr><td>CES AC</td><td>HEATER</td><td></td></tr><tr><td>CES DC</td><td>ECs</td><td></td></tr><tr><td>LGC</td><td>GLYCOL(ON IF TEMP</td><td></td></tr><tr><td>RCS A REG</td><td>>50°)</td><td></td></tr><tr><td>RCS B REG</td><td></td><td></td></tr></table> | WARN | CAUT | COMP | ASC PRESS(ON THRU DES) | PREAMP | H2O SEP | CES AC | HEATER | | CES DC | ECs | | LGC | GLYCOL(ON IF TEMP | | RCS A REG | >50°) | | RCS B REG | | |
| WARN | CAUT | COMP | | | | | | | | | | | | | | | | | | | | | |
| ASC PRESS(ON THRU DES) | PREAMP | H2O SEP | | | | | | | | | | | | | | | | | | | | | |
| CES AC | HEATER | | | | | | | | | | | | | | | | | | | | | | |
| CES DC | ECs | | | | | | | | | | | | | | | | | | | | | | |
| LGC | GLYCOL(ON IF TEMP | | | | | | | | | | | | | | | | | | | | | | |
| RCS A REG | >50°) | | | | | | | | | | | | | | | | | | | | | | |
| RCS B REG | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | | | | |
| | CB(16) LTG: ANUN/DOCK/COMPT - Close
STAB/CONT: ATCA - Close
HEATER: DISP - Close | | | | | | | | | | | | | | | | | | | | | | |
| | CB(11) STAB/CONT: ENG CONT - Close | | | | | | | | | | | | | | | | | | | | | | |
| 2 | RCS SYS A/B-2: QUADS(4) - AUTO
HTR TEMP MONITOR - Cycle Then LDG
(HEATER Lt - Off)
LAMP/TONE TEST - Check All Positions | | | | | | | | | | | | | | | | | | | | | | |
| 3 | PRIM EVAP FLOW No 1 - OPEN
GET _____: _____ | | | | | | | | | | | | | | | | | | | | | | |
| 4 | Close CB's Per ACTIVATION PWR UP Chart | | | | | | | | | | | | | | | | | | | | | | |

C/W CHECKOUT

ACT-23

ACTIVATION PWR UP

16

FLT DISP		ECS SYS 8				PROPUL				ASC No				
EVENT NAME /	ASC	ASC	WOL	QUAD 1	QUAD 2	QUAD 3	QUAD 4	COMP	TEMP/PRESS	PODS/	MAIN	DISP/IND	PODS	ASC No
BY TIME	FEED 1	FEED 2	TYPE	TYPE	TYPE	TYPE	TYPE		TEMP/PRESS	PODS/	MAIN	DISP/IND	PODS	FEED

4

LITG		ED		STAB/CONT				INST				ECS		
FLOOD	TRACE	AMPLITUDE/	WAVE	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC
		LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC
		LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC

3

COMM		ECS		ECS		ECS		ECS		ECS		ECS		ECS	
DISP	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC	LOC

4

HEATERS		CAME		ECS		ECS		ECS		ECS		ECS		ECS	
QUAD 1	QUAD 2	QUAD 3	QUAD 4	QUAD 5	QUAD 6	QUAD 7	QUAD 8	QUAD 9	QUAD 10	QUAD 11	QUAD 12	QUAD 13	QUAD 14	QUAD 15	QUAD 16

4

TB VERIFICATION

ACT-24

97:10

TB VERIFICATION

- 1 CB(16) INST: CMEA - Open Then Close
Cycle TEMP MON

WARN	CAUT	COMP
ASC PRESS	ECS	SUIT FAN
RCS A REG		H2O SEP
RCS B REG		
- 2 FUEL & OXID VENT (2) -tb-gray
LDG GEAR DEPLOY - tb-bp
- 3 ASCENT He REG 1&2 -tb-gray
DESCENT He REG 1-tb-gray
DESCENT He REG 2 -tb-bp
- 4 SYS A&B ASC FUEL & OXID (4)-tb-bp
SYS A&B QUADS (8) - tb-gray
CRSFD tb-bp
SYS A&B MAIN SOV -tb-gray
- 5 RECORDER - OFF - tb-bp

Basic Date 2/6/70
Changed

ACT-25

97:12

PGNS TURN-ON & SELF TEST

- 1 Check Bus Voltages
- 2 V35E
F 88 88
(Master Alarm, LGC & ISS Warning,
And All DSKY Lts - On,
8's In All Registers; Lts
Reset In 5 sec)
- 3 CB(11) PGNS: IMU OPR - Close
NO ATT Lt - On (Off In 90 sec)
Wait 20 sec After NO ATT Lt - Off,
then
V37E00E
- 4 V25 N01E 1365E
E,E,E
- 5 V15 N01E 1365E
R1,R2,R3 All Zero

ACT-26

6 V21 N27E 10E (Test
Fixed And Erasable Memory)

R1 Number Of Errors
R2 Number Of Tests Started
R3 Number Of Tests Successful
(Test Successful If $R2 > 3$ Within
78 sec)

```
*
*PROG Lt-On
*
* V05 N09E 01102 SELF-
* TEST ERROR
* N08E Record For MSFN
*
* R1
*
* R2
*
* R3
*
```

7 V21 N27E 0E TERMINATE SELF TEST

LM 7

Basic Date 2/6/70
Changed 3/16/70
ACT 27

***** UD - 2:00 (97:16) *****

***** AOS 97:16 *****

Report LM Pwr Transfer Time (ACT-17)

97:16

* SEC S-BAND VOICE CHECK

Notify MSFN of SEC S-BD CK
Perform SEC S-BD VOICE CK With MSFN
(Up To 60 sec To Lock)

97:18

* PRIM S-BAND T/R AND PWR AMPL CHECK

- 1 Notify MSFN of PRIM S-BD CK
S-BAND XMTR/RCVR - PRIM
S-BAND PWR AMPL - PRIM
(Up To 60 sec To Relock)

- 2 Perform Comm Check With MSFN

ACT-28

97:21

97:21

ECS ACTIVATION & CHECKOUT

* S-BAND STEERABLE ANTENNA ACTIVATION

- 1 02/H20 QTY MON - ASC 2, ASC 1, DES 1 HTR CONT TEMP MONITOR - S-BAND
(-52° to +135°)
- 2 SUIT ISOL (2) - SUIT FLOW
SUIT ISOL (2)-ACTUATE OVRD (Suit Disc)
SUIT GAS DIVERTER - PUSH/CABIN
S-BAND -PM, PRIM, VOICE, PCM,
RANGE, OFF, HI
CSM Mnvr To Proper Attitude
- 3 SUIT FAN - 2 (Master Alarm (Twice),
SUIT/FAN Warning Lt-On &
SUIT FAN Comp Lt-On
Momentarily, ECS Caution,
H20 SEP Comp Lts - ON
Then Off In 2 Min)
- 2 HI GAIN: PITCH - -75°
YAW - -12°
TRACK MODE - SLEW (Wait 30 sec)
PITCH (From MSFN) (+112)CCW
YAW (From MSFN) (+ 34)CCW
ANTENNA S-BAND - SLEW
- 3 Verify Signal Strength > 3.0
TRACK MODE - AUTO (>4.0)
- 4 S-BAND CHECK WITH MSFN

Basic Date _____ 2/6/70
Changed _____ 3/16/70
ACT 29

97:23

97:23

CDR CONNECT TO LM ECS

LMP CONNECT TO LM ECS

- 1
- Connect To CDR Hoses
(Stow Gas Connector Plugs In Purse)
SUIT ISOL - SUIT FLOW
Verify (192 PKG) Lanyard Stopper Not
Seated
CB(16) ECS: LCG PUMP - Close
PRESS REG A - EGRESS (Suit Gas Diverter
Automatically Extends)

Disconnect CSM 02 Hose & Return Hose to CSM
Connect To LMP Hoses
SUIT ISOL - SUIT FLOW

ACT 30

97:26

DOCKED IMU COARSE ALIGN

- 1 Verify CSM In Min DEADBAND ATT HOLD 1 CB(16) ECS: SUIT FAN 2 - Open
(Master Alarm, SUIT/FAN Warning
SUIT FAN Comp Lts - On)
- 2 Calculate LM Gimbal Angles

OG	IG	MG
<u>300.00</u>	<u>180.00</u>	<u>360.00</u>
_____ +Rc (See TLC-1)		
(<u>7.5</u>)	(<u>112.5</u>)	(<u>22.5</u>)
-CM	+CM	-CM

LM	LM	LM
(<u>292.5</u>)	(<u>292.5</u>)	(<u>337.5</u>)

- 3 V41 N20E COARSE ALIGN IMU
F 21 22 LOAD ICDU ANGLES OG,IG,MG (.01°)
(NO ATT Lt - On, FDAI Torques) 1
*PROG Lt-On *
*V05 N09E 00211 COARSE *
* ALIGN ERROR,Go* *
* To 3 *

SUIT FAN/H2O SEP CHECK

97:26

GLYCOL PUMP CHECK

97:28

- CB(11) ECS: GLYCOL PUMP 1 - Open
(Master Alarm, ECS Caution
Lt - On Momentarily)
- CB(11) ECS: GLYCOL PUMP 1 - Close
(GLYCOL Comp Lt-On)

LM 7

Basic Date 2/6/70
Changed

ACT-31

- 4 V40 N20E ZERO CDU (NO ATT Lt-Off) 2 GLYCOL - INST (SEC) (8 psia)
Notify CSM ATT HOLD No Longer Required CB(16) ECS: GLYCOL PUMP SEC - Close
(10-20 psi Rise)
: GLYCOL PUMP SEC - Open
- 5 V25 N07E
F 21 07 SET REFSMFLG 3 GLYCOL - PUMP 2 (21-37 psi)
77E, 10000E, 1E, V01 N01E, 77E Confirm (GLYCOL Comp Lt - On Then Off)
Bit 13 Is Set (Set If 1st Digit Is CB(11) ECS: GLYCOL PUMP AUTO
1, 3, 5, or 7) TRNFR-Open
GLYCOL - PUMP 1 (21-37 psi)
- 6 V37E 51E
PRO 4 Biomed Sw - Right
V37E 00E
- 7 V06 N20 On LM MARK - ENTR
Note Time; Copy CSM & LM OG, IG, MG
GET : :
OG CM IG MG
CM LM LM
LM LM LM
- 8 Voice Gimbal Angles And Time To MSFN
AUDIO (Both): VHF B - T/R
- 2 Both CDR & LMP Perform Voice Check
On VHF Simplex B

97:30
VHF B CHECKOUT

1 CSM Configure for VHF Simplex B
VHF B XMTR - VOICE
VHF B RCVR - ON
VHF ANT - FWD

VHF C/O
LGC/CMC CLOCK SYNC

ACT 32

*****UD - 1:45 (97:31)*****

97:34

VHF A CHECKOUT

1 CSM Configure For VHF Simplex A

VHF A XMTR - VOICE

VHF A RCVR - ON

VHF B XMTR - OFF

AUDIO (Both): VHF B - RCV

: VHF A - T/R

2 Both CDR & LMP Perform Voice Check On
VHF Simplex A

97:36

*LGC/CMC CLOCK SYNC/TEPHEM UPDATE

1 V25 N36E

2 Load CSM Time : : :

3 On CSM Mark - ENTR

4 V06 N65, On Mark - ENTR
Compare With CSM N65

CSM Time : : :

LM Time : : :

V55E - Load Δ T
Check Mission Timer

Basic Date 2/6/70
 Changed
 ACT 33

97:38

5 Record CSM TEPHEM

R1

R2

R3

ASCENT BATTERY ACTIVATION & CHECKOUT

1 CB(16) EPS: ASC ECA CONT - Close

2 POWER/TEMP MON SEL - BAT 5
 BAT 5 NORMAL FEED-ON (Verify BAT
 Current)

3 POWER/TEMP MON SEL - SE BUS Then BAT 6
 BAT 6 NORMAL FEED-ON (Verify BAT
 Current)

4 BAT 1,2 HI-VOLT-OFF/RESET
 BAT 3,4 HI-VOLT-OFF/RESET
 Verify BAT Current = 0
 POWER/TEMP MON SEL-CDR BUS Then SE BUS

5 BAT 5 BACKUP FEED-ON
 BAT 6 BACKUP FEED-ON
 BAT 5 NORMAL FEED-OFF/RESET
 BAT 6 NORMAL FEED-OFF/RESET
 POWER/TEMP MON SEL-CDR BUS, SE BUS, Then
 BAT Current

6 V25 N01E, 1706E Load TEPHEM (Octal)

7 V05 N01E, 1706E Verify TEPHEM

8 Verify MSFN Contact
 V74E (Erasable Dump) (42 Sec)

DAP SET

ASC BAT CK

14:46

SET DAP

1 V48E
R1 32022
PRO

2	F 06 47	LM, CSM	wt. (lbs)
	R1	_____	(33731)
	R2	_____	(37500)
	PRO		

```

3 F 06 48 GMBL TRIM, PITCH, ROLL (.01°)
      R1 _____ (+00476)
      R2 _____ (+00572)
      (TERM) V34E_____

```

6 BAT 1&2 HI VOLT-ON
BAT 3&4 HI VOLT-ON
Verify BAT Current

```

7 BAT 5 BACKUP FEED-OFF/RESET
  BAT 6 BACKUP FEED-OFF/RESET
    Verify BAT Current = 0

```

8 CB(16) EPS: ASC ECA CONT - Open

9 Record ED BAT Voltage For MSFN

BAT A

BAT B

Basic Date _____ 2/6/70
Changed _____
ACT-35

97:43

LANDING GEAR DEPLOY

- 1 CB(11) ED: LDG GEAR FLAG-Close
: LOGIC POWER A-Open
MASTER ARM-ON
LDG GEAR DEPLOY-FIRE, tb-gray
CB(11) ED: LOGIC POWER A-Close
LDG GEAR DEPLOY-FIRE
MASTER ARM-OFF
CB(11) ED: LDG GEAR FLAG-Open

97:45

RCS PRESSURIZATION

- 1 RECYCLE: SYS A&B ASC FEED 2(2) - CLOSE
SYS A&B ASC FEED 1(2) - OPEN
- 2 RCS QUANTITY A&B - 100%
SYS A&B ASC FUEL & ASC OXID - tb(4) Remain-bp
SYS A&B THRUSTER PAIR QUADS - tb(8) gray
(Possible tb-Red, Cycle CMEA If Necessary)
RECYCLE: CRSFD-CLOSE
: MAIN SOV SYS A&B - OPEN
HTR CONT TEMP MON - Check RCS QUADS (>120°)

LDG GEAR DEPLOY
RCS PRESS

LDG GEAR DEPLOY
RCS PRESS

ACT 36

- 3 TEMP/PRESS MON - He (2820-3280 psia)
PRPLNT (40°-100°/10-50 psi)
FUEL MANF (25-90 psi)
OXID MANF (25-90 psi)
- 4 CB(16) LOGIC PWR B-Open
MASTER ARM - ON
HE PRESS RCS - FIRE
(RCS A&B REG Warning Lts-Off)
RECYCLE: SYS A&B ASC FEED 2(2) - CLOSE
CB(16) LOGIC PWR B-Close
MASTER ARM-OFF
- 5 RECYCLE: SYS A&B ASC FEED 1(2) - OPEN
: SYS A&B THR PAIR QUADS(8)-OPEN
: CRSFD - CLOSE
: SYS A&B MAIN SOV-OPEN
- 6 TEMP/PRESS MON - OXID MANF (175-188 psi)
- FUEL MANF (175-188 psi)
- PRPLNT (40°-100°/178-188 psi)
- He (2750-3200 psi)
Read He Pressure To MSFN

***** UD - 1:30 (97:46) *****

Basic Date _____ 2/6/70
Changed _____
ACT 37

97:49

*RCS CHECKOUT

- 1 GUID CONT - PGNS
ATT TRANSL - 4 JET
ATT CONT (3) - PULSE
MODE CONT (Both) - ATT HOLD
ACA/4 JET (CDR) - DISABLE
TTCA (CDR) - JETS
Verify HBR With MSFN & CSM In
Wide Deadband & Attitude Hold
QUAD Flags - Red & RCS TCA Lt - on will
occur during cold fire checks

2 TTCA (Cold Fire) Check

V76E
V11N10E, 5E
CDR TTCA
UP (+X) - R1 00252
DN (-X) - 00125
E, 6E
RIGHT (+Y) - R1 00220
LEFT (-Y) - 00140
FWD (+Z) - 00011
AFT (-Z) - 00006

RCS C/O

ACT 38

3 PGNS RATE CMD (Cold Fire), AGS PULSE (Cold Fire) Check

CB(11) ATT DIR CONT - CLOSE

V77E

V15 NOTE, 42E

CDR ACA (To Soft Stop, Pause 2 sec At Null)

ROLL RIGHT R3 00045-00057

ROLL LEFT 77720-77732

PITCH UP R1 00045-00057

PITCH DN 77720-77732

YAW RIGHT R2 77720-77732

YAW LEFT 00045-00057

4 AGS RATE CMD (Cold Fire), 4 JET SEC

COIL (Hot Fire) Check

Verify CMC MODE - FREE

GUID CONT - AGS

ATT CONT (3) - MODE CONT

ACA/4 JET (CDR) - ENABLE

CDR ACA (Deflect Slowly To Hardover, Pause 2 sec At Null)

ROLL - RIGHT

ROLL - LEFT

PITCH - UP

PITCH - DN

YAW - RIGHT

YAW - LEFT

LM 7

Basic Date 2/6/70

Changed

Basic Date 2/6/70
Changed

ACT 39

5 PGNS MIN IMP (Hot Fire) Check
GUID CONT - PGNS
V76E

CB(11) RCS SYS A: QUAD TCA (4) - Close
CB(16) RCS SYS B: QUAD TCA (4) - Close
CB(16) INST: CWEA - Open Then Close
CYCLE TEMP MON

V11N10E, 31E R1 67777

CDR ACA (Out Of Detent (2 1/2°), Pause 2 sec At Null)

ROLL RIGHT - R1 27757

ROLL LEFT - R1 27737

YAW RIGHT (Twice) - R1 27767

YAW LEFT (Twice) - R1 27773

V48E, V21E, 31022E, PRO, PRO, V34E

V11N10E, 31E

CDR ACA (Out of Detent (2 1/2°), Pause 2 sec At Null)

PITCH UP - R1 27776

PITCH DN - R1 27775

Notify CSM Hot Fire Checks Complete

6 ATT/TRANSL - 2 JET
V37E 00E

***** SS 97:59 *****

97:59

97:59

*IMU FINE ALIGN

*MSFN UPLINK/UPDATE

1 Copy Ground Calculated Gyro
Torquing Angles

1 UPDATE LINK - DATA
MSFN P-27 Updates LS REFSMMAT,
LM STATE VECTOR And LGC Abort Constants
UPDATE LINK - OFF

X _____, Y _____, Z _____

2 V76E (Verify)
V42E Fine Align IMU
F 21 93 Load Gyro Torquing
Angles X,Y,Z (.001°)

2 Copy Updates
Gyro Torquing Angles
AGS Abort Constants
DAP Data

3 V16 N93E Monitor Torquing
(All Zero)

***** UD - 1:15 (98:01) *****

Basic Date 2/6/70
 Changed 3/16/70

ACT-41

98:01AGS ACTIVATION AND SELF TEST

- 1 AGS STATUS - STBY (Master Alarm,
 AGS Warning Lt-On)
 CB(16) STAB/CONT: AEA-Close
 (AGS Warning Lt-Off)
 CB(11) AC BUS B: AGS - Close
 AGS STATUS - OPERATE
 (Master Alarm & AGS Warning Lt-On)
 02/H20 QTY MON-C/W RESET

2 000+888888 (OPR ERR Lt-On)

3 123-45679

98:04DROGUE AND PROBE INSTALLATION

1

Verify:

Both Electrical Umbilicals Removed
 Drogue Lock Lever Engaged & Flush
 Three Capture Latches Engaged & Locked
 LM Hatch Exterior Insulation O.K.
 Flaps Secured Around Handles

2

Close & Secure Hatch
 CABIN DUMP (OVHD) - AUTO
 PRESS REG A&B - CABIN
 SUIT GAS DIVERter - PUSH/CABIN

4 412+0 REINITIATE TEST

412R +1 SELF TEST SATISFACTORY

+3 LOGIC TEST FAILURE

+4 MEMORY TEST FAILURE

+7 LOGIC AND MEMORY TEST FAILURE

5 574R DESCENT STAGE FLAG (+ Not Staged)

6 604R LUNAR SURFACE FLAG (+ NOT On

Lunar Surface)

7 612R STAGING SEQ COUNTER (+0 Nom)

DROGUE & PROBE

AGS T/O

ACT-42

98:07

AGS TIME INITIALIZATION

1 V16 N65E
Set AGS Time (377) 90 hr Bias

98:09

LOAD AGS PAD

1 224 _____ (+60427)
225 _____ (+29402)
226 _____ (+60469)
305 _____ (-01718)
662 _____ (-54500)
673 _____ (-31701)

2 232 +00600
233 +00250
464 +00500

Basic Date _____ 2/6/70
Changed _____ 3/16/70

Basic Date _____ 2/6/70
Changed _____

ACT-43

465 +00195

616 +0

623 +0

514 R _____ (-60000)

515 R _____ (-44223)

516 R _____ (+00000)

3 Copy AGS K FACTOR Update

_____ : _____ : _____

V47E

N16 GET OF AGS CLOCK

V25E LOAD AGS K FACTOR UPDATE

V34E

***** UD - 1:00 (98:16) *****

98:19

1 Match Indicated Angles
 TRACK MODE - SLEW,
 Set P (+130)
 Y (+ 40)
 S-BD ANT - AFT
 VHF B XMTR - DATA
 BIOMED-OFF, PCM-LO
 UPLINK SQUELCH - ENABLE

***** LOS (98:21) *****

98:21

Don Helmet & Gloves

98:21

Don Helmet & Gloves

Basic Date 2/6/70

Changed

Basic Date 2/6/70Changed

ACT 45

98:28PGA PRESSURE INTEGRITY CHECK

- 1 SUIT GAS DIVERter - PULL/EGRESS
CABIN GAS RETURN - EGRESS
SUIT CIRCUIT RELIEF - CLOSE
PRESS REG A - CLOSE
PRESS REG B - DIRECT 02 (Suit Press to
8.85 psia)
PRESS REG B - CLOSE (Monitor Cuff Gage,
Decay <.3 Psi in 1 min)
- 2 C02 CANISTER SEL - SECONDARY (C02 Comp
Lt-On, Monitor Cuff Gage, <.3 psi In
1 min)
C02 CANISTER SEL - PRIMARY (C02 Comp
Lt-Off)
- 3 SUIT CIRCUIT RELIEF - AUTO
PRESS REG A&B - CABIN
CABIN GAS RETURN - AUTO
SUIT GAS DIVERter - PUSH/CABIN

***** UD -45 (98:31) *****PGA CK
REG CK

ACT-46

98:33

REGULATOR CHECK

- 1 Verify CSM Tunnel Hatch, Press Equalization,
And Tunnel Vent vlvs Closed, And Tunnel Vented
- 2 CABIN GAS RETURN - EGRESS
Verify: OVHD CABIN DUMP VALVE - AUTO
: CABIN REPRESS - AUTO
PRESS REG A&B - EGRESS
(SUIT GAS DIVERter - EGRESS)
READ STEP 3 BEFORE PROCEEDING
- 3 FWD CABIN DUMP VALVE - OPEN Then AUTO At
Master Alarm, CABIN Warning Lt - On
Verify AUTO CABIN REPRESS Between 4.45 to 3.7 psi

As Soon As Possible:
CB (16) ECS: CABIN REPRESS - Open
(CABIN Warning Lt - Off, Cabin
Repress Stops)
- 4 FWD CABIN DUMP VALVE - OPEN Then AUTO
At 3.5 psi (Verify Suit Press 3.6
to 4.3 psi)

Basic Date 2/6/70
Changed

Basic Date _____ 2/6/70
Changed _____

ACT-47

- 5 PRESS REG A&B - CLOSE
SUIT CIRCUIT RELIEF - OPEN
At Suit Press of Approximately 3.5 psi,
SUIT CIRCUIT RELIEF-AUTO
PRESS REG B - EGRESS (Suit Press
3.6 to 4.0 psi)
- 6 PRESS REG A&B - CABIN
CB (16) ECS: CABIN REPRESS - CLOSE
(CABIN Warning Lt - On,
REPRESS Vlv Opens)
Cabin Press Rises 4.6 to 5.0 psia
(CABIN Warning Lt-Off)
CABIN GAS RETURN - AUTO
SUIT GAS DIVERter - PUSH/CABIN
DOFF HELMET & GLOVES (CREW OPT)

DRIFT CK
RATE GYRO CK

ACT-48

98:43

98:43

DRIFT CHECK

RATE GYRO CHECK

V06N20 On LM MARK - ENTR

1

Verify CSM Holding Attitude
GYRO TEST - POS RT (RPY RATE +5°/sec)
GYRO TEST - NEG RT (YPR RATE -5°/sec)

GET : : :

OG

IG

MG

2

RATE SCALE -5° SEC
REPEAT Tests

. CM . CM . CM

. LM . LM . LM

(Will Transmit Angles To MSFN At AOS)

***** UD - :30 (98:46) *****

***** SR 98:48 *****

LM 7

Basic Date 2/6/70
Changed

Basic Date 2/6/70Changed

ACT-49

98:4998:49RNDZ RDR SELF TESTAGS UPDATE & ALIGN

- 1 CB(11) RR(2) - Close (NO TRACK Lt-0n) 1 TLM-HI
Verify: CSM RCS Thruster B3 - Off V47E, 414+1
: Radar Xponder - Off
- RNDZ RDR ANT - Released 2 400 + 3 AGS/PGNS Align
- X-POINTERS (Both) - HI MULT
- RATE/ERR MON (Both) - RNDZ RADAR 3 V83E, 317R, 440R
- ATTITUDE MON (Both) - PGNS
- MODE SEL - LDG RDR 4 TLM-L0

- 2 RNG/ALT MON - RNG/RNG RATE
SHFT/TRUN - +50°
RR MODE - SLEW
TEMP MONITOR - RNDZ (+10° To +50°)

98:54AGS CALIBRATION

- CB(11) AC BUS A: RNG/RNG RT/ALT/ALT
RT - Close 1 V16 N20E
- RR GYRO SEL - SEC
FLIGHT DISPLAYS: RNG/RNG RT/ALT/ALT
RT-Close 16 20 ICDU Angles, 0, I, M
CSM Mnvr Until LM ICDU'S:
292.5 (OG)
292.5 (IG)
337.5 (MG)
RATES <.075°/sec In All Axes
- 2 V40 N20E ICDU ZERO

RR SELF TEST
AGS CAL

RR SELF TEST
AGS CAL

ACT-50

3 SLEW RATE-HI

Slew Left To Mode I Region (18 sec)
Slew Right, Down, Left, Up
(FDAI Needles Right, Down, Left, Up)
SLEW RATE - LO
SHFT/TRUN - + 5°
Slew Right, Down, Left, Up
(FDAI Needles Right, Down, Left, Up,
1°/sec; X-Pointer-3 mr/sec)

4 RR MODE - AUTO TRACK

RADAR TEST - RNDZ RDR (Rng Rt Tape
Drives To -473 to -515 fps, X-Pointers
and FDAI Needles Vary Between +5°.
After 12 sec Rng Tape Drives to
194.19 to 196.99NM, NO TRACK Lt-Off)

5 TEST MONITOR - AGC (1.6)

- XMTR (2.4)
- SHAFT ERR (2.0 To 2.8
@1/2cps)
- TRUN ERR (2.1 To 2.7
@1/2cps)
- AGC

3 Read and record: ACCEL BIAS COEFF

540 X _____ (.001 ft/sec 2)
(-00002)

541 Y _____ (.001 ft/sec 2)
(+00001)

542 Z _____ (.001 ft/sec 2)
(-00002)

GYRO DRIFT COEFF

544R X _____ (.01/hr)
(-00006)

545 Y _____ (.01°/hr)
(+00030)

546R Z _____ (.01°/hr)
(+00047)

4 Verify CSM Thrusters Disabled

LM 7 Basic Date 2/6/70
Changed 3/16/70

ACT-51

- 5 400+6 CALIBRATE GYRO & ACCEL
After 32 sec:
Read and Record
540R (.001 ft/sec 2)
541R (.001 ft/sec 2)
542R (.001 ft/sec 2)
Values Should Not Change From Step 3
By More Than .039 ft/sec 2 (.008nom)
- 6 400R (+0 After 302 sec)
Notify CSM To Enable A11 Thrusters
Except B3 (B3 Can Be Enabled If
RR SELF TEST Complete)
Read and Record
544R (.01°/hr)
545R (.01°/hr)
546R (.01°/hr)
Values Should Not Change From
Step 3 By More Than 2.0°/hr
(.9 Nominal)
- 6 Set NORRMON Flag
V25 N07E
101E, 10E, 1E
RR MODE - LGC (NO TRACK Lt - On)
Wait 10 sec)
- 7 V63E Start RR Self Test
F 04 12
R1 00004 Specify Radar
R2 00001 Rndz Radar
PRO
NO TRACK Lt-On (Off After 12 sec)
- 8 F 16 72 TRUN, SHAFT (.01°)
R1 Varying At 1/2 cps
R2 Varying at 1/2 cps
PRO
RR SELF TEST
AGS CAL
- 9 F 16 78 RANGE, RANGE RATE, TFI (.01nm,
fps,min-sec)
R1 +195.39 To +195.79 (TM Within +1.2
of R1)
R2 -00475 To -00517 (TM=2<R2)
V34E
- 10
- 11 RADAR TEST -OFF (NO TRACK Lt-On,
X-Pntr-Center)

12 V40 N72E RR CDU ZERO (10 sec)

SHFT/TRUN - +50°

13 V41 N72E (+04000, +04000)

PRO

V16N72E

V44E

14 SHFT/TRUN -+5°

RR GYRO SEL - PRIM

V41 N72E (+35600, +35600)

PRO

V16N72E

V44E

15 V41 N72E (+00000, +28300)

PRO

V16N72E

CB(11) RR(2) - Open

(NO TRACK Lt-Off)

V44E

Notify CSM That Thruster B3-Off, And
Radar Xponder-Off Are No Longer Required

16 RATE/ERR MON (LMP)-LDG RDR/CMPTR

ATT MON (LMP) - AGS

***** UD - :15 (99:01) *****

Basic Date 2/6/70
 Changed

ACT-53

99:01

PREP FOR UNDOCKING

- 1 Verify Undocking Attitude (0, XXX/282,060)
 S-BD-PM, PRIM, PRIM, VOICE,
 PCM, RANGE
 VHF-VOICE, ON, DATA, ON, RIGHT, LO
 AUDIO (Both): VHF A-T/R
 : VHF B-RCV

- 2 MISSION TIMER-SET
 EVENT TIMER-SET, Count Up to 99:16:21 (Undocking)
 OVHD HATCH-LOCKED
 OVHD CABIN RELIEF & DUMP - AUTO
 PRESS REG A&B - CABIN

- 3 GUID CONT - PGNS
 MODE SEL - LDG RADAR
 RNG/ALT MON - RNG/RNG RT
 RATE ERR MON (CDR) - RNDZ RDR
 (LMP) - LDG RDR/CMPTR
 ATTITUDE MON (CDR) - PGNS
 (LMP) - AGS
 RATE SCALE - 5°/SEC

PREP FOR UNDOCK

ACT-54

- 4 ATT/TRANSL - 2 JET
 BAL CPL - ON
 DEADBAND - MAX
 ATTITUDE CONTROL (3) MODE CONT
 MODE CONT (Both) - ATT HOLD
 TTCA (Both) - JET
 RR MODE - SLEW
 CB(11) HTRS: AOT - Close
 Mount Camera On Window Bar
 LM 3 /DAC/10/CEX-ULC
 (fTT, 250, ∞) 6 fps, .06 Mag (1 min)
 LM /DC/60/HCEX
 (fTT, 250, focus) 10
 Mount TIMELINE Book
- 5 Configure CB Panels Per UNDOCKING Chart

ACT-55

50542

ACT-56

UNDOCKING

16

[illegible][illegible][illegible][illegible]

NASA — MSC

LM 7

Basic Date: 2/6/70
Changed: _____

Basic Date 2/6/70
 Changed 3/16/70
 ACT-57

UD - : 10 (99:06)

6 CHECK ATTITUDE (0, 150/282, 060)
 V62E

7 V48E
 R1 21002
 PRO

F 06 47 LM, CSM Wt. (1bs)
 R1 (33731)
 R2 (37500)
 PRO

F 06 48 GMBL TRIM, PITCH, ROLL (.01)
 R1 (+00476)
 R2 (+00572)
 (TERM) V34E

HELMET AND GLOVES ON

ACT-58

AOS 99:10

99:10

S-BD ANT-AFT, Verify Comm
CK S-BD P (+130)
 Y (+ 40)

S-BD ANT - SLEW (>3.0)
TRACK MODE - AUTO (>4.0)
VHF B XMTR - OFF
BIOMED - LEFT, PCM-HI
UPLINK SQUELCH - OFF

***** GO/NO GO FOR UNDOCKING *****

Voice DRIFT CHECK Gimbal Angles
& Time To MSFN
(SEE ACT-48)

LM 7

Basic Date 2/6/70
Changed

NASA — MSC

Basic Date 2/6/70
Changed

ACT-59

8 TAPE RECORDER - ON

9 P47

404 + 0E

405 + 0E

406 + 0E

470R

Insert V77 (DO NOT ENTR)

99:16:21

***** UNDOCKING *****

Go To LM TIMELINE BOOK

